

EngChain Taxonomy

1 Chemical Engineering

**Transport Phenomena**
Shell Momentum Balances
Viscosity & Momentum Transport

**Reaction Kinetics**
Mole Balances
Stoichiometry
Conversion & Reactor Sizing

**Thermodynamics**
Heat Effects
Volumetric Properties of Pure Fluids

2 Electrical Engineering

**Signals & Systems**
Discrete Time Signals
Continuous Time Signals

**Electromagnetics & Waves**
Electrostatics
Magnetostatics
Waves & Phasors

**Digital Communications**
Deterministic & Random Signal Analysis
Digital Modulation Schemes

3 Civil Engineering

**Structural Analysis**
Determinate Structures
Influence Lines
Deflections
Indeterminate Analysis

**Geotechnical Engineering**
Phase & Index Properties
Permeability, Seepage & Effective Stress
Stress Distribution & Consolidation
Strength & Stability

**Water Resources**
Uniform Flow
Energy & Rapidly Varied Flow
Hydrology

4 Industrial Engineering

**Production & Inventory**
Deterministic Lot Sizing
Stochastic Inventory
Production Planning

**Quality & Reliability Control**
Variables Control Charts
Attributes Control Charts
Process Capability
Acceptance Sampling

**Stochastic Operations**
Queueing Systems
Markov Chains
Poisson Processes
System Reliability

5 Mechanical Engineering

**Fluid Mechanics**
Fluid Statics
Fluid Kinematics

**Mechanics of Materials**
Torsion
Stress & Strain

**Vibrations & Acoustics**
Harmonically Excited Vibrations
Single Degree of Freedom Systems